HTML & CSS CASE STUDY REPORT

A report explaining the main topics in three website projects

Case Study 1: Personal Portfolio Website

Case Study 2: NRI University Website

Case Study 3: PULSEFIT Gym Website

Prepared from the six supplied HTML and CSS files.

1. Introduction

This report explains three website projects created with HTML and CSS. HTML provides the page structure and content, while CSS controls presentation such as colors, spacing, typography, layout, hover effects, and responsiveness. Each project links its HTML file to a separate stylesheet.

2. Technologies and Common Concepts

HTML (HyperText Markup Language): Uses elements such as <header>, <nav>, <section>, , <table>, <form>, and <footer> to organize the content.

CSS (Cascading Style Sheets): Styles elements using selectors such as element selectors (body, header), class selectors (.card, .container), ID selectors (#about), pseudo-classes (:hover, :focus), and media queries (@media).

External stylesheet: Each HTML page connects to a CSS file using a link element in the head. This keeps content separate from design and makes styling easier to maintain.

Navigation with anchor links: Links such as href="#skills" or href="#programs" jump to matching sections with the corresponding id.

Responsive design: CSS media queries and flexible layouts adjust content for smaller screens. Flexbox and CSS Grid are used in the university and gym websites.

3. Case Study 1 — Personal Portfolio Website

Purpose: A simple portfolio page introducing a B.Tech CSE Core student and presenting academic and personal information.

Main HTML sections

- **Header:** Displays the student name and course description.
- **Navigation bar:** Links to Profile, Education, Skills, Hobbies, Projects, Achievements, Resume, and Contact.
- **Profile:** Brief introduction and interest in software development.
- **Education:** Lists B.Tech CSE Core, Intermediate, and school details.
- **Technical skills:** Python, HTML, CSS, and C programming.
- **Hobbies and projects:** Gaming, movies, learning technology, a family tree website, and Games Hub.
- **Achievements:** College-level chess and participation in an Atal Tinkering Lab motherboard activity.
- **Resume and contact:** Includes a print button using window.print(), email, location, and a footer.

Main CSS topics

- Arial font, light-blue page background, centered text, and a navy header/footer.
- White sections with margins, padding, and rounded corners.
- Navigation hover color changes to yellow; sections gain a shadow on hover.
- A media query at 600px adjusts navigation link spacing for narrow screens.

Key learning: Semantic sections, ID-based navigation, external CSS, hover effects, and a simple print action.

4. Case Study 2 — NRI University Website

Purpose: A college information website presenting institutional details and key student-related information.

Main HTML sections

- **Header and banner:** Shows the university name, tagline, welcome message, and call to action text.
- **Navigation:** Links to Home, About, Departments, Courses, Faculty, Admissions, Facilities, Events, and Contact.
- **About:** Brief description of education, innovation, research, and student development.
- **Departments and courses:** Uses card-like blocks to list CSE, ECE, Mechanical, Civil, B.Tech courses, and MBA.
- **Faculty:** Displays example names and roles.
- **Admissions:** Shows a five-step process from checking eligibility to confirming admission.
- **Facilities and events:** Lists library, hostel, labs, sports, cafeteria, Wi-Fi, technical fest, cultural fest, and sports meet.
- **Contact and footer:** Displays address, email, phone number, and copyright text.

Main CSS topics

- A light background with dark-blue header/footer and blue navigation.
- A banner with pale-blue background and generous spacing.
- Flexbox-based .container layout with wrapping, centered cards, and gaps.
- Cards have white backgrounds, padding, rounded corners, and shadows.
- A media query at 600px changes navigation links and makes cards wider on small screens.

Key learning: Building a multi-section institutional page, reusable card layouts, Flexbox, and basic responsive styling.

5. Case Study 3 — PULSEFIT Gym Website

Purpose: A gym website promoting workout programs, facilities, trainers, class schedules, membership plans, and registration.

Main HTML sections

- **Header/navigation:** Brand logo and links to Programs, Facilities, Trainers, Schedule, Pricing, and Register.
- **Hero section:** A bold slogan, short description, and Join The Gym link.
- **Workout programs:** Heavy Strength, HIIT Conditioning, Endurance & Engine, and Mobility & Flow.
- **Facilities:** Lists lifting platforms, dumbbells, turf, recovery facilities, cardio equipment, boxing studio, lockers, and access.
- **Trainer profiles:** Cards with initials, names, specialties, and short descriptions.
- **Class schedule:** An HTML table organized by time and selected days.
- **Pricing:** Basic, Pro Pass, and Elite membership cards, including a highlighted popular plan.
- **Registration form:** Collects name, email, phone, preferred plan, and fitness goal using required inputs and select fields.
- **Footer:** Displays the gym copyright statement.

Main CSS topics

- CSS variables in :root for colors and font family; a dark theme with lime accent color.
- Universal selector and box-sizing reset, smooth scrolling, typography, and reusable .container.
- Flexbox for navigation and facility items; CSS Grid for program, trainer, facility, and pricing cards.
- Hover transitions on buttons and cards, sticky header, highlighted popular pricing card, and form focus styling.
- Media query at 768px hides the navigation links and adjusts heading/form sizes for smaller screens.

Important implementation note: The form prevents normal submission and shows an alert. It is a front-end demonstration; no server-side storage or real membership processing is included in the supplied code.

Key learning: CSS variables, Grid, Flexbox, tables, forms, hover transitions, and responsive layout.

6. Comparison of the Three Projects

Feature	Portfolio	University	Gym
Main purpose	Personal introduction	Institutional information	Gym promotion and sign-up
Layout approach	Simple stacked sections	Flexbox cards	Grid cards and table
Interactive styling	Hover effects, print button	Card hover and responsive rules	Hover transitions, form focus
Responsive rule	600px breakpoint	600px breakpoint	768px breakpoint

7. Conclusion

The three projects demonstrate how HTML and CSS can be used for different purposes: a personal portfolio, a university information site, and a gym promotional site. Across the projects, the main concepts include semantic HTML structure, navigation links, external stylesheets, CSS selectors, spacing, color and typography, hover effects, responsive media queries, and reusable layouts. The PULSEFIT project additionally demonstrates CSS variables, Grid, a schedule table, pricing cards, and a registration form.

Note: This report describes the supplied code as written. It does not claim that the university details, contact information, or gym registration are connected to live services.